

Course Name

Semester / Year

Your Name

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Chapter 1

Introduction

1.1 Getting Started

Note:-

This template uses the **Tokyo Night Dark** colour scheme. Replace placeholder text with your own notes.

Definition 1.1.1: Example Definition

Write your definition here. You can use maths: $f : \mathbb{R} \rightarrow \mathbb{R}$.

Theorem 1.1.1 Example Theorem

State your theorem here. For example, $\forall \epsilon > 0 \exists \delta > 0$ such that...

Proof: Write your proof here.



Example 1.1.1 (Example)

Show an example here.

Note:-

Add more chapters and sections as needed. Happy note-taking!

1.2 Code Snippets

Note:-

Inline code: use `\py{x = [i**2 for i in range(10)]}`, `\cpp{int x = 0;}`, `\java{System.out.println();}`, or the generic `\inlinecode{...}` for any language.

```
1 def fibonacci(n: int) -> list[int]:
2     """Return the first n Fibonacci numbers."""
3     a, b = 0, 1
4     result = []
5     for _ in range(n):
6         result.append(a)
7         a, b = b, a + b
8     return result
9
10 print(fibonacci(10))
```

Listing 1.1: Python example

```

1 #include <iostream>
2 #include <vector>
3
4 int main() {
5     std::vector<int> v = {1, 2, 3, 4, 5};
6     for (auto x : v) {
7         std::cout << x << " ";
8     }
9     return 0;
10 }

```

Listing 1.2: C++ example

```

1 import java.util.List;
2 import java.util.ArrayList;
3
4 public class Main {
5     public static void main(String[] args) {
6         List<String> names = new ArrayList<>();
7         names.add("Alice");
8         names.add("Bob");
9         names.forEach(System.out::println);
10    }
11 }

```

Listing 1.3: Java example

LAYER
1 —
Explain
Like I'm
5

Explanation Like I am five Imagine you have a box of crayons. Each colour represents a different idea...

LAYER
2 —
The
Real
Expla-
nation

Formally, let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function such that...

LAYER
3 —
Key
Take-
aways

- Continuity preserves connectedness
- The Intermediate Value Theorem applies

MISCONCEPTION
ALERT

Many students think that “continuous” means “differentiable”. This is **false**: $|x|$ is continuous at 0 but not differentiable there.

OPTIONAL

Further Exploration

See Rudin, *Principles of Mathematical Analysis*, Chapter 4 for a deeper treatment.

Core Idea

The authors propose a novel attention mechanism that reduces quadratic complexity to linear.

Key Arguments / Findings

- Sparse attention patterns capture long-range dependencies
- Empirical gains on GLUE and long-document benchmarks

Methodology

Experiments on 5 datasets, ablation studies, comparison to Transformer, Reformer, and Longformer.

Contributions

1. First linear-time attention with provable approximation guarantees
2. Open-source implementation integrated with HuggingFace

Limitations / Criticism

- Assumes causal masking; bidirectional variant not explored
- Hyperparameter sensitivity in low-resource settings

Discussion Points

How might this approach interact with mixture-of-experts routing? Could it be adapted for multimodal inputs?

Discussion Points

This is test

Discussion Points

This is test two

Discussion Points

this is test